Practical 1 : Source Code and Output

Code :-

```
# ======================================================
# TABLE-DRIVEN VACUUM CLEANER AGENT
# Every important line is commented.
# ======================================================

# Environment representation.
environment = {'A':'Dirty', 'B':'Dirty'}

# Agent starts in Room A.
location='A'

# Action table.
# Key = (Current Room, Room Status)
# Value = Action
action_table={
    ('A','Dirty'):'SUCK',
    ('A','Clean'):'MOVE_RIGHT',
    ('B','Dirty'):'SUCK',
    ('B','Clean'):'MOVE_LEFT'
}

# Repeat until goal state.
while True:

    # Display current state.
    print("Location:",location)
    print("Environment:",environment)

    # Goal test.
    if environment['A']=='Clean' and environment['B']=='Clean':
        print("Action: STOP")
        break

    # Create current percept.
    percept=(location,environment[location])

    # Lookup action.
    action=action_table[percept]

    print("Selected Action:",action)

    # Execute action.
    if action=='SUCK':
        environment[location]='Clean'
```

```
    elif action=='MOVE_RIGHT':
        location='B'

    elif action=='MOVE_LEFT':
        location='A'

    print("-"*40)

print("\nAnalysis")
print("The table-driven agent always chooses an action")
print("based on the current percept stored in the table.")
print("Hence it behaves rationally in this environment.")
```

Output :-

